ADC Hardware

ECE 2564, Embedded Systems

Fall 2025

ADC Hardware

- Flash Converter
- Simle Count-up ADC
- Successive Approximation Register (SAR) ADC

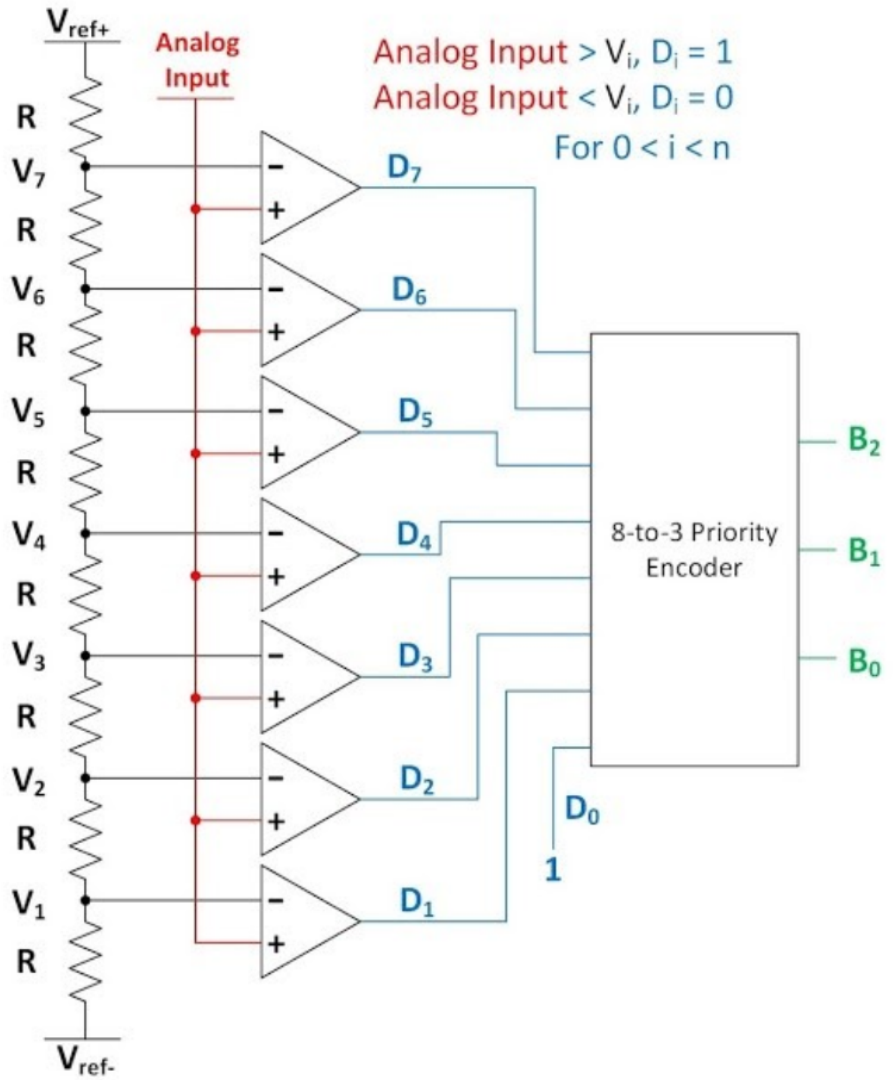
- Pros and Cons

Flash Converter for $n = 3$

$$V_{LSB} = (V_{ref+} - V_{ref-}) / 8$$

$$V_i = V_{ref-} + i * IR$$
$$= V_{ref-} + i * V_{LSB}$$

$I = V_{LSB} / R$



Flash Converter for $n = 3$ and EFSR $= 4\,\mathrm{V}$

$V_{LSB} = (V_{ref+} - V_{ref-}) / 8$

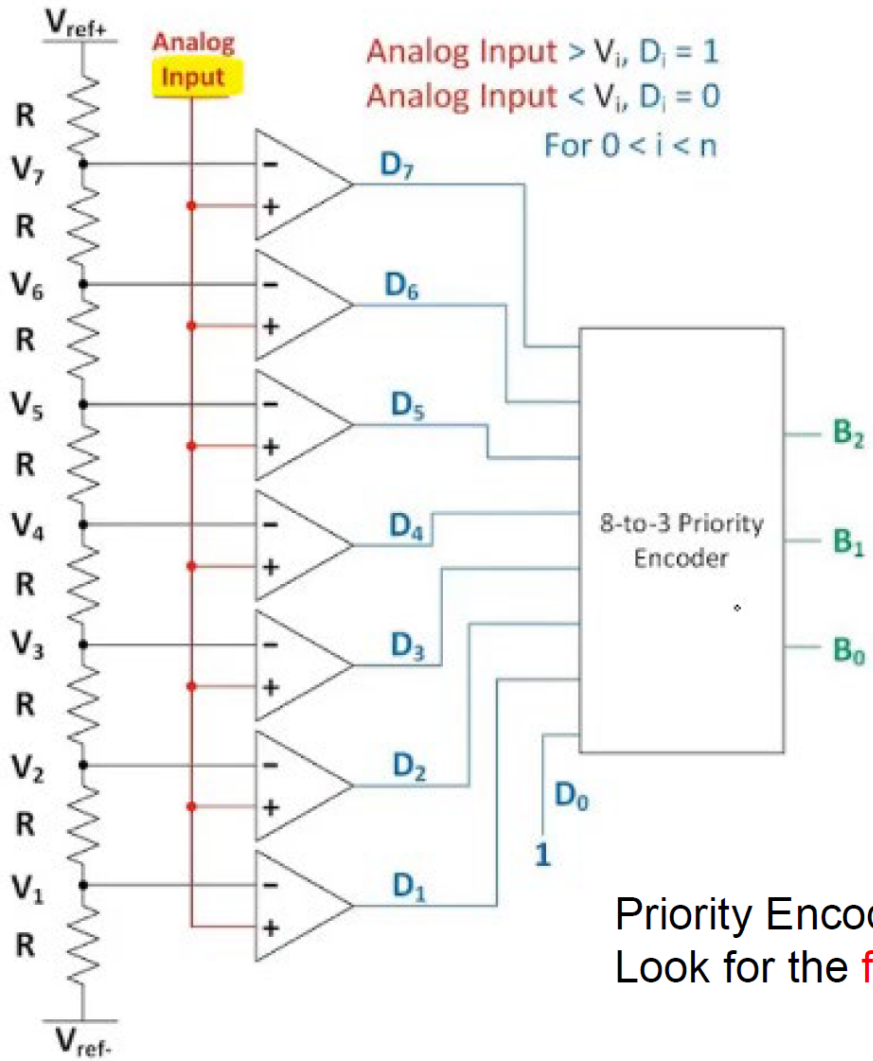
$V_i = V_{ref-} + i * IR$
$= V_{ref-} + i * V_{LSB}$

$I = V_{LSB} / R$

Vref- = 0
Vref+ = 4V

EFSR = 4 - 0 = 4
1LSB = EFSR / 2^3 = 4/8 = 0.5

V2 = Vref- + 2(0.5)

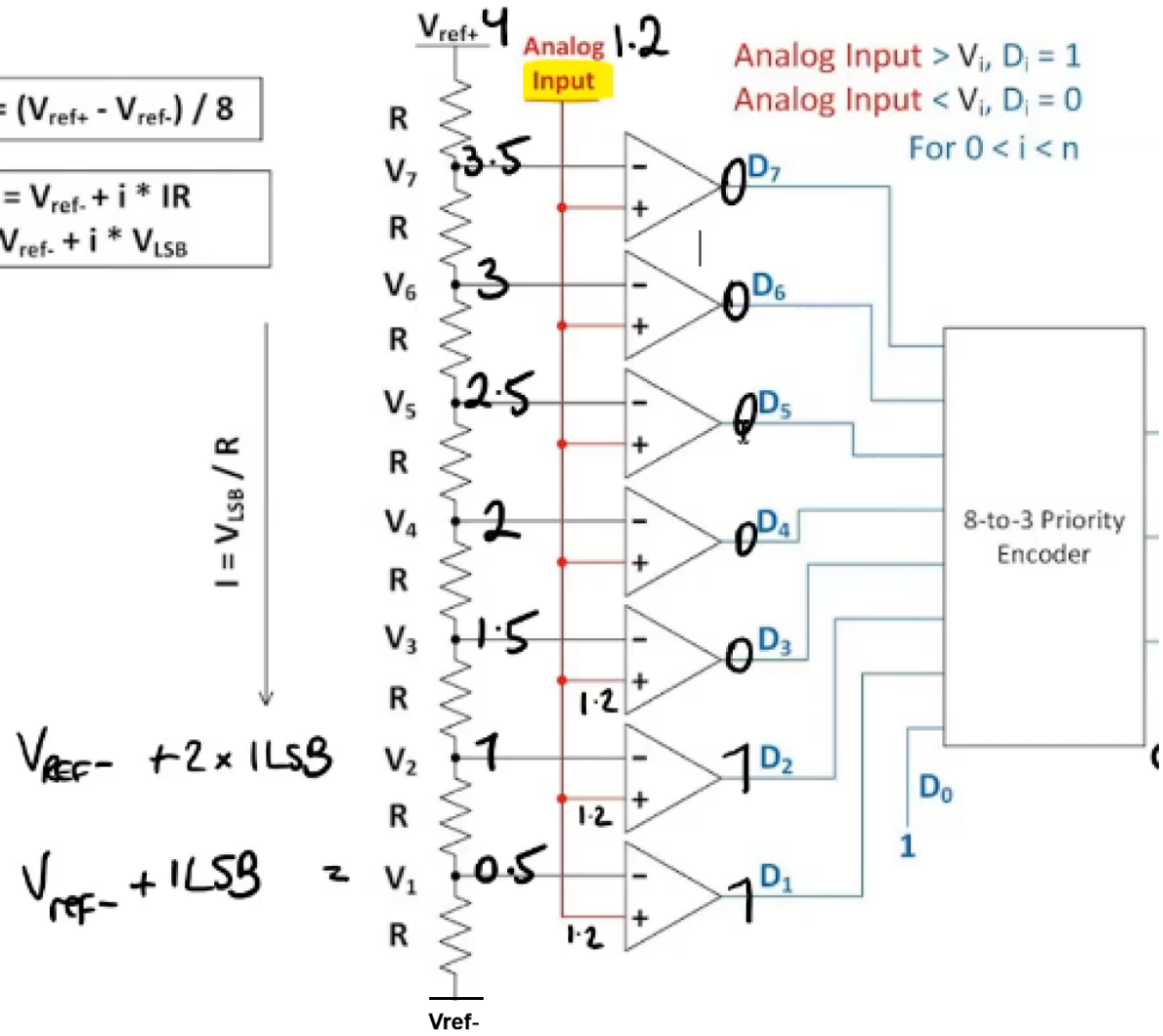


Priority Encoder:
Look for the first 1

Example: Vin $= 1.2\,\mathrm{V}$

$$V_{LSB} = (V_{ref+} - V_{ref-}) / 8$$

$$V_i = V_{ref-} + i * IR$$
$$= V_{ref-} + i * V_{LSB}$$



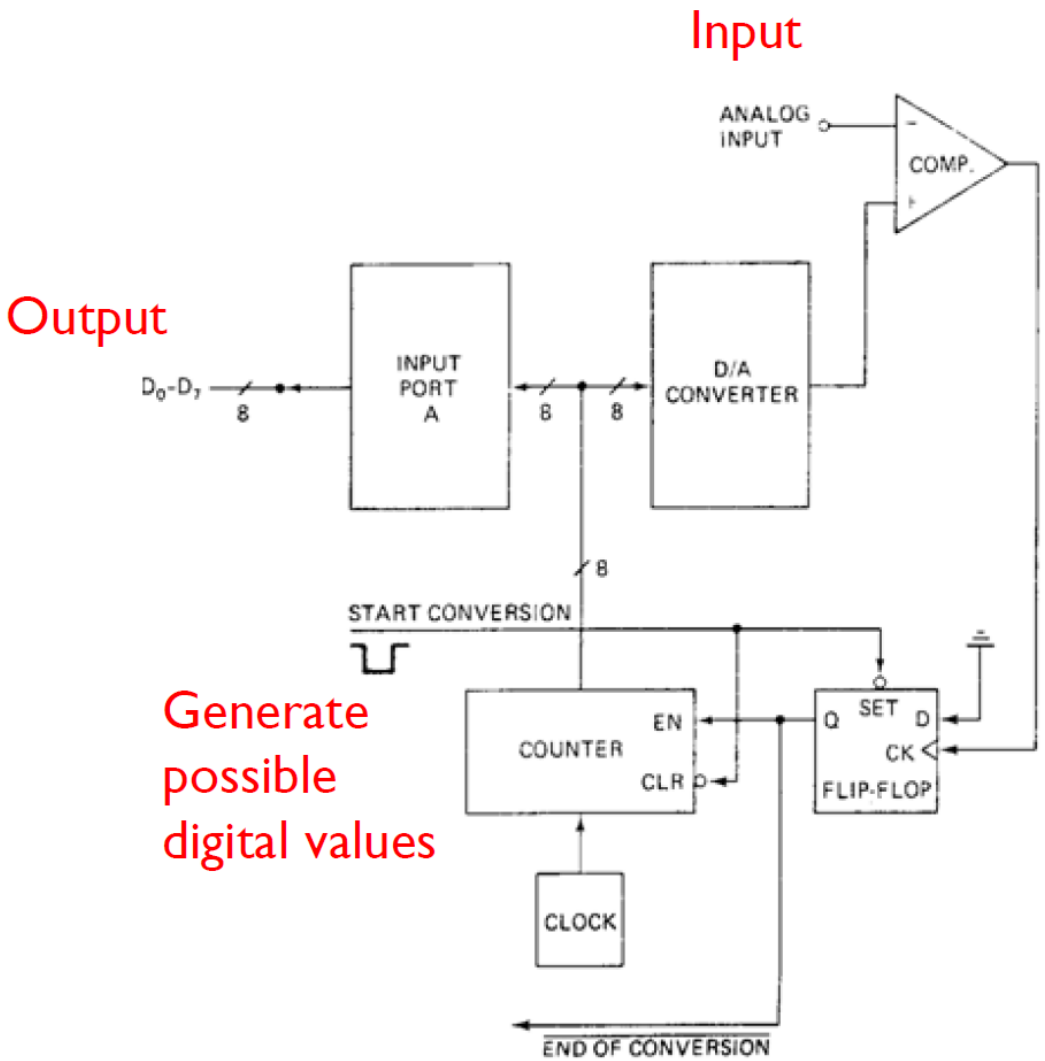
- D_0 = 1
- Op-amps: 0000011
- Input of Encoder: 00000111
- Output of Encoder: 010

Flash Converter

- Op-amps serve as comparators
  - $2^n - 1$ op-amps, serving as comparators, since $D_0 = 1$
- Resistors provide reference input values to the op-amps
  - 2^n resistors
- Very *fast* conversion

- What if $n = 14$?
  - We would need $2^14 - 1$ op-amps!

Counting Converter Built From a DAC!



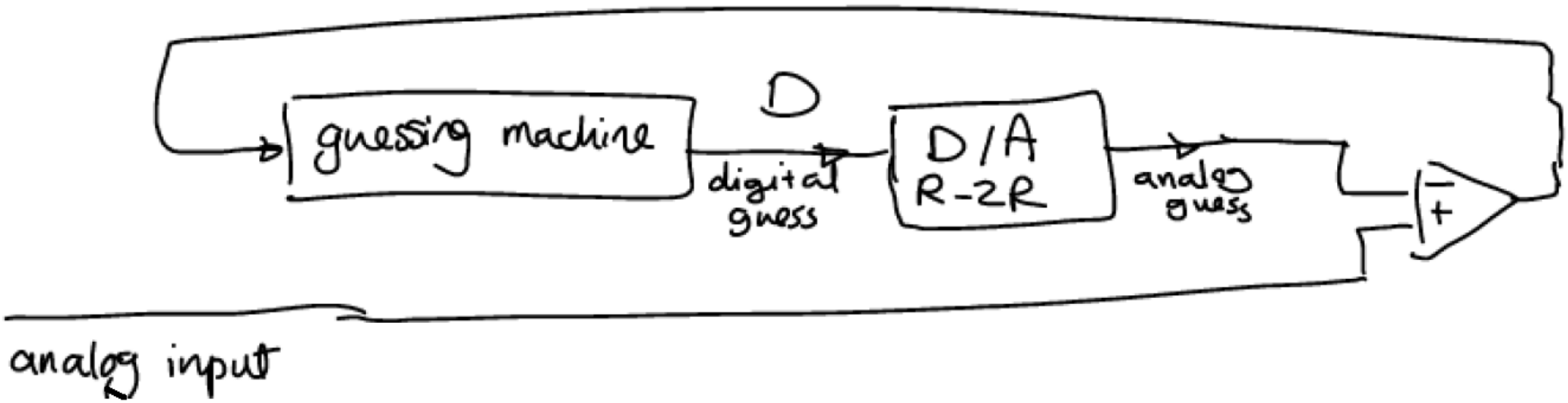
Counting Converter

- Use an n-bit counter to produce a candidate binary input to a DAC
- Compare the DAC output to the ADC input
- Increment the counter value until the DAC out matches (or exceeds) the ADC input

- This is *very slow*

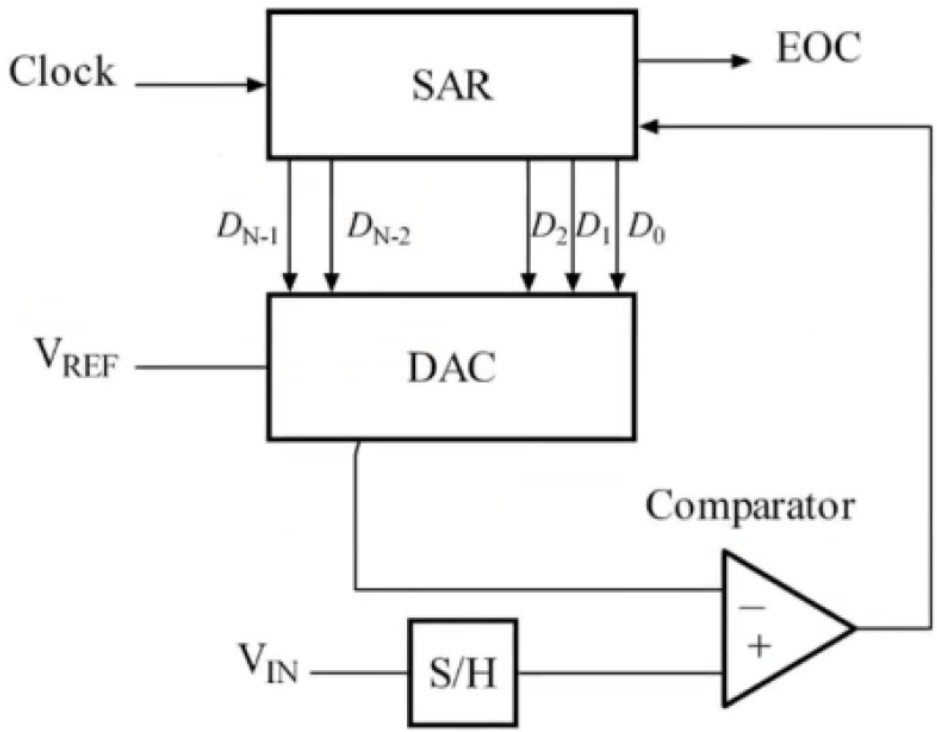
Simple Count-Up Converter

- We introduce a "guessing machine" to guess the digital equivalent of the the input signal's voltage
- How should the guess be made?
  - Start at 0 and increment by 1?
  - This is what we just saw on the last page
  - On average we would need about 2^{n-1} guesses
  - The larger the analog signal the longer it would take to generate the answer (the output voltage)



Successive Approximation Register (SAR) ADC

- Similar to the simple count-up method, except the "guessing" is more sophisticated



Successive Approximation Register ADC

A Better Guessing Machine: Binary Search

- Average number of guesses is n for an n-bit ADC
- Example: Input is 85, $n = 7$ with a range from 0 to 127
- Guess 0: $0 = 0\text{b}0000000$
- Guess 1: $0 + 2^6 = 64 = 0\text{b}1000000$
- Guess 2: $64 + 2^5 = 64 + 32 = 96 = 0\text{b}1100000$
- Guess 3: $96 - 2^4 = 96 - 16 = 80 = 0\text{b}1010000$
- Guess 4: $80 + 2^3 = 80 + 8 = 88 = 0\text{b}1011000$
- Guess 5: $88 - 2^2 = 88 - 4 = 84 = 0\text{b}1010100$
- Guess 6: $84 + 2^1 = 84 + 2 = 86 = 0\text{b}1010110$
- Guess 7: $86 - 2^0 = 86 - 1 = 85 = 0\text{b}1010101$

- At most $\log_2 2^7 = 7$ guesses are needed
  - Much better than 85 guesses for the simple count-up

SAR ADC

- Key idea: At the $(i+1)$-th iteration, the i-th most significant bit is determined
- Example $V_{\text{ref}} = 4\,\text{V}$; $V_{\text{in}} = 1.2\,\text{V}$; $n = 3$

- Guess 1: $0\text{b}100$; Is $2\,\text{V} > 1.2\,\text{V}$? Yes
- Guess 2: $0\text{b}010$; Is $1\,\text{V} > 1.2\,\text{V}$? No
- Guess 3: $0\text{b}011$; Is $1.5\,\text{V} > 1.2\,\text{V}$? Yes

- Final Output: $0\text{b}010$
  - NOTE: Sometimes you might not flip the final guess

Summary

There is a lot going on when analog-to-digital conversion is performed

- Signal conditioning: noise reduction and amplification
- Nyquist criterion and sampling rate calculation
- ADC hardware
  - Often implemented with a DAC!